

LAB 17.1

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Task 1: Customer Data Cleaning

Prompt

Clean customer dataset using Pandas by handling missing values, duplicates, and encoding.

Code

```
import pandas as pd

df =
    pd.DataFrame({ 'age':
        [20, None, 25],
        'income': [50000, 60000, None],
        'city': ['Chennai', 'DELHI', 'Mumbai'],
        'gender': ['Male', 'Female', 'Male']
    })

df.fillna(df.mean(numeric_only=True), inplace=True)
df.drop_duplicates(inplace=True)

df['city'] = df['city'].str.lower()

df['gender'] = df['gender'].map({'Male':0, 'Female':1})
df['city'] = df['city'].astype('category').cat.codes

print(df)
```

Output

	age	income	city	gender
0	20.0	50000.0	0	0
1	22.5	60000.0	1	1
2	25.0	55000.0	2	0

Task 2: Sales Preprocessing

Prompt

Preprocess sales data by converting dates, creating features, and normalizing values.

Code

```
import pandas as pd

df = pd.DataFrame({
    'date': ['2024-01-01', '2024-02-01'],
    'sales': [1000, 2000]
})

df['date'] = pd.to_datetime(df['date'])
df['month'] = df['date'].dt.month
df['year'] = df['date'].dt.year

df['sales'] = (df['sales'] - df['sales'].min()) / (df['sales'].max() - df['sales'].min())

print(df)
```

Output

	date	sales	month	year
0	2024-01-01	0.0	1	2024
1	2024-02-01	1.0	2	2024

Task 3: Healthcare Data

Prompt

Clean and preprocess healthcare dataset with scaling and train-test split.

Code

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import pandas as pd

df =
    pd.DataFrame({ 'bp':
        [120, None, 140],
        'chol': [200, 220, None],
        'gender': ['M', 'F', 'M']
    })

df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df['gender'] = df['gender'].map({'M':0,'F':1})

scaler = StandardScaler()
df[['bp','chol']] = scaler.fit_transform(df[['bp','chol']])

train, test = train_test_split(df, test_size=0.2)

print(train)
```

Task 4: Employee Salary

Prompt

Preprocess employee salary dataset by removing outliers and encoding categories.

Code

```
import pandas as pd

df =
    pd.DataFrame({ 'salary':[30000,40000,
        1000000],
        'department':['HR','IT','IT'],
        'location':['Hyd','Chennai','Delhi']
    })

df['department'] = df['department'].str.lower()

df = df[df['salary'] < 500000]

df['department'] = df['department'].astype('category').cat.codes
df['location'] = df['location'].astype('category').cat.codes

print(df)
```

Output

	salary	department	location
0	30000	0	1
1	40000	1	0